ME-2 Digital Twin Baseline Construction

Week 8: Combined Static and Dynamic Response Study

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1 Purpose

The purpose of this study is to analyze the combined static and dynamic behavior of the instrumented beam system under simultaneous loading and vibration excitation. Earlier stages of the project examined static load-response behavior and dynamic vibration response separately. This stage combines both effects into a single synchronized experiment.

The experimental workflow uses three primary implementation files:

- `load_cell_test.ino`
- `data.csv`
- `sensor_test.m`

The Arduino implementation acquires the experimental measurements, the CSV dataset stores the synchronized sensor signals, and the MATLAB script performs the signal processing and frequency-domain analysis presented throughout this report.

The overall objective is to establish a measurable behavioral baseline that can later be used by the digital twin for expected-response prediction and anomaly detection.

2 Experimental Architecture

The experimental system consists of a 6061 aluminum beam instrumented with two load cells and a triaxial accelerometer. A DC vibration motor provides controlled excitation through a PWM signal generated by the Arduino-based acquisition system.

The experimental workflow is summarized below:

1. `load_cell_test.ino` acquires synchronized sensor measurements

2. Experimental measurements are streamed and stored in `data.csv`

3. `sensor_test.m` imports and processes the dataset

4. MATLAB computes time-domain and frequency-domain response metrics

5. Processed results are visualized and interpreted

This architecture creates a complete sensing-to-analysis pipeline for the behavioral digital twin.

3 Embedded Data Acquisition Implementation

The embedded acquisition system is implemented in:

`load_cell_test.ino`

This Arduino-based implementation interfaces with:

- HX711 load-cell amplifiers
- ADXL345 accelerometer
- PWM-controlled vibration motor
- Serial output logging

The embedded implementation continuously records:

- Load Cell 1 response
- Load Cell 2 response
- Vibration motor magnitude
- Accelerometer x response
- Accelerometer y response
- Accelerometer z response

The Arduino code outputs synchronized measurements at approximately:

$$\Delta t = 0.01 \text{ s}$$

which corresponds to:

$$F_s = 100 \text{ Hz}$$

sampling frequency.

This timing relationship later becomes important during FFT computation in the MATLAB implementation.

4 Experimental Dataset Structure

The synchronized measurements generated by the Arduino implementation are stored in:

`data.csv`

The dataset contains six numerical columns with no column headers.

Table 1: Experimental Dataset Structure

Column	**Signal**	**Description**
Column 1	Load Cell 1	Primary structural load response
Column 2	Load Cell 2	Secondary structural load response
Column 3	Vibration Magnitude	PWM-driven excitation magnitude
Column 4	Accelerometer X	x-direction acceleration
Column 5	Accelerometer Y	y-direction acceleration
Column 6	Accelerometer Z	z-direction acceleration

Each row represents one synchronized measurement instant from the experiment.

The MATLAB implementation imports the dataset directly and separates the columns into dedicated processing variables.

5 MATLAB Processing Workflow

The signal processing and visualization implementation is contained in:

`sensor_test.m`

The script performs the following operations:

1. Imports the CSV dataset

2. Extracts synchronized sensor signals
3. Removes invalid measurements
4. Constructs a physical time vector
5. Computes resultant acceleration magnitude
6. Computes statistical response metrics
7. Performs FFT analysis
8. Identifies dominant vibration frequency
9. Generates visualization plots

The MATLAB implementation forms the computational portion of the behavioral digital twin pipeline.

6 CSV Import and Signal Extraction

The MATLAB implementation first imports the dataset using:

`readcell(fileName)`

The script then separates the imported dataset into individual signal vectors:

$$LC1 = \texttt{data(:,1)}$$
$$LC2 = \texttt{data(:,2)}$$
$$Vibe = \texttt{data(:,3)}$$
$$Xacc = \texttt{data(:,4)}$$
$$Yacc = \texttt{data(:,5)}$$
$$Zacc = \texttt{data(:,6)}$$

Rows containing invalid measurements are removed so that all remaining signals remain synchronized throughout processing.

This directly connects the physical experimental measurements stored in `data.csv` to the computational variables used later in the MATLAB implementation.

7 Time Vector Construction

The MATLAB implementation converts the sample index into a physical time scale using:

$$t_i = i\Delta t$$

where:

- t_i is physical time
- i is the sample index
- $\Delta t = 0.01$ s

This relationship is implemented in the MATLAB code as:

```
t = (0:N-1)' * dt
```

This conversion allows the experimental response to be analyzed in actual physical time rather than sample number.

8 Combined Time-Domain Response

Figure 1 shows the measured combined-response behavior.

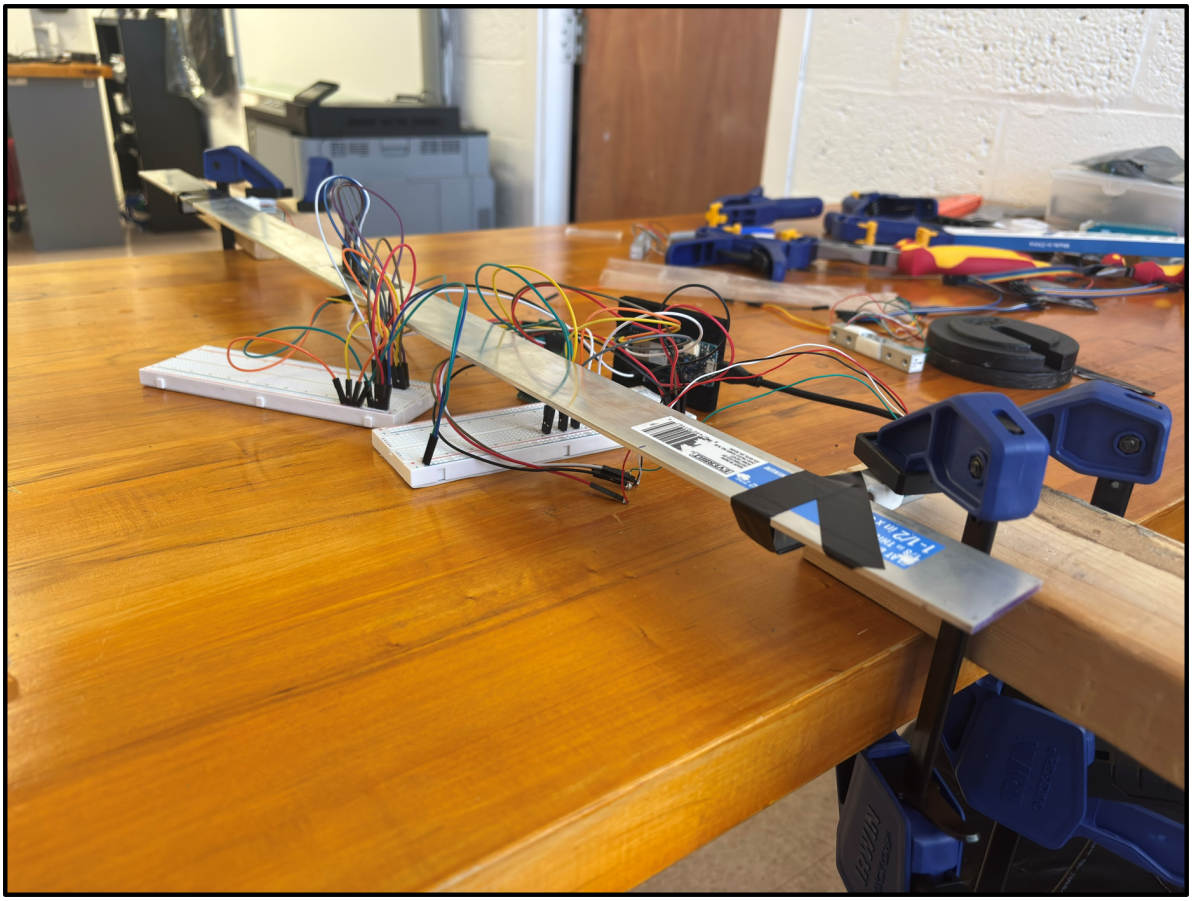


Figure 1: Combined static and dynamic time-domain response.

The load-cell signals show static loading behavior while the accelerometer signals show continuous dynamic excitation caused by the vibration motor.

Several important response phases are visible:

- Baseline response region
- Load application region
- Load redistribution region
- Load removal region

The difference between Load Cell 1 and Load Cell 2 indicates that the applied force was distributed unevenly across the structure.

This demonstrates that the experiment captures both magnitude and spatial variation of the loading condition.

9 Resultant Acceleration Magnitude

The accelerometer measures vibration along three orthogonal directions:

$$a_x, \quad a_y, \quad a_z$$

The MATLAB implementation combines these signals into a single resultant acceleration magnitude using:

$$a_{\text{mag}} = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

This equation is implemented directly in:

`sensor_test.m`

as:

```
AccelMag = sqrt(Xacc.^2 + Yacc.^2 + Zacc.^2);
```

This resultant magnitude represents the overall vibration intensity independent of accelerometer orientation.

10 Time-Domain Metrics

Several statistical metrics are computed from the resultant acceleration signal.

The RMS acceleration is computed using:

$$a_{\text{RMS}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} a_i^2}$$

This operation is implemented in MATLAB as:

```
aRMS = rms(AccelMag);
```

The standard deviation is computed as:

$$\sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(a_i - \bar{a})^2}$$

and implemented using:

```
aStd = std(AccelMag);
```

The tolerance band is then defined as:

$$\text{tol} = 2\sigma$$

which appears in the MATLAB implementation as:

```
tol = kTol * aStd;
```

These equations connect the measured vibration behavior directly to the response metrics computed in the processing script.

11 Frequency-Domain Analysis

The MATLAB implementation performs frequency-domain analysis using the Fast Fourier Transform.

Before applying the FFT, the mean value is removed from the signal:

$$a_{\text{detrended}} = a_{\text{mag}} - \bar{a}$$

This operation isolates the oscillatory vibration behavior and removes the DC offset.

The FFT is then computed using:

$$Y = \texttt{fft}(a_{\text{detrended}})$$

which is implemented directly in the MATLAB script as:

```
Yfft = fft(aDetrend);
```

The single-sided FFT magnitude spectrum is then computed and plotted.

12 Frequency-Domain Response

Figure 2 shows the FFT magnitude spectrum generated from the resultant acceleration signal.

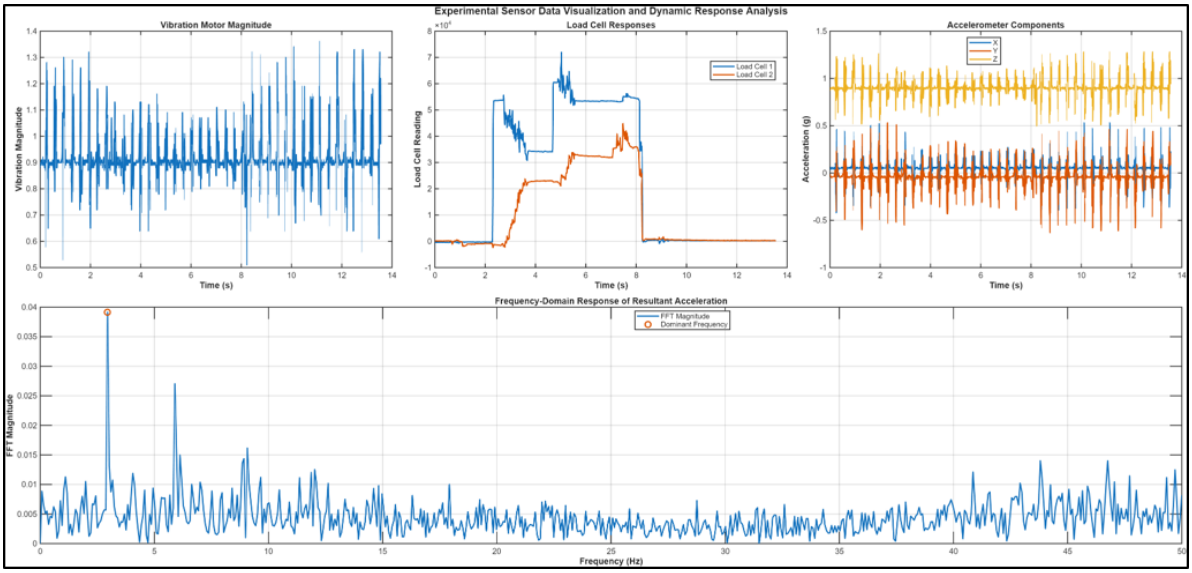


Figure 2: FFT magnitude spectrum of the resultant acceleration response.

The dominant frequency is identified by locating the maximum FFT magnitude:

$$f_{\text{dom}} = f(\text{argmax}(P_1))$$

This relationship is implemented in MATLAB using:

```
[peakMag, peakIdx] = max(P1);
```

followed by:

```
domFreq = f(peakIdx);
```

The dominant response frequency was approximately:

$$f_{\text{dom}} \approx 2.95 \text{ Hz}$$

This frequency represents the strongest measured vibration component during the experiment.

13 Combined Static and Dynamic Interpretation

The experiment demonstrates that the structure exhibits coupled static and dynamic behavior during loading.

The load cells capture force redistribution while the accelerometer captures continuous vibration response. The simultaneous measurement of both behaviors is important because

real structures often experience loading and vibration together rather than independently.

The results suggest that changing load conditions may slightly influence the dynamic response characteristics of the beam.

This is significant for the behavioral digital twin because it shows that structural response cannot always be represented using isolated static or dynamic models alone.

14 Digital Twin Relevance

The combined-response dataset provides a behavioral baseline for future digital twin operation.

The current implementation establishes:

- Expected load-cell response behavior
- Expected acceleration response behavior
- Baseline FFT frequency characteristics
- Coupled static-dynamic response behavior
- Statistical response envelopes

These baseline relationships can later be used for residual-based anomaly detection.

A general digital twin residual can be written as:

$$r(t) = y_{\text{measured}}(t) - y_{\text{predicted}}(t)$$

The measured signals originate directly from:

`data.csv`

while the predicted behavior is generated computationally using the MATLAB processing implementation.

15 Summary

Week 8 combines static loading response and dynamic vibration response into a single synchronized experimental study.

The workflow connects:

- Embedded sensing and acquisition
- Experimental dataset generation

- MATLAB signal processing
- Statistical response characterization
- FFT-based frequency analysis

The study demonstrates that the instrumented beam system can successfully capture coupled structural behavior and establishes a measurable behavioral baseline for future digital twin development.

A Appendix A: Associated Project Files

The following files were used throughout this study and have been uploaded to the project GitHub repository for documentation, reproducibility, and future development.

Table 2: Associated Project Files

File Name	Purpose
`load_cell_test.ino`	Arduino-based embedded acquisition implementation used for synchronized collection of load-cell, vibration, and accelerometer measurements.
`data.csv`	Experimental dataset containing synchronized sensor measurements collected during the combined static and dynamic response experiment.
`sensor_test.m`	MATLAB processing implementation used for dataset import, signal processing, FFT analysis, metric computation, and visualization generation.

B Appendix B: GitHub Repository Contents

The GitHub repository associated with this project contains:

- Embedded Arduino acquisition code
- MATLAB processing and visualization scripts
- Experimental datasets
- Generated plots and figures
- Weekly digital twin deliverables

The repository serves as the primary location for project version control, reproducibility, and continued digital twin development.